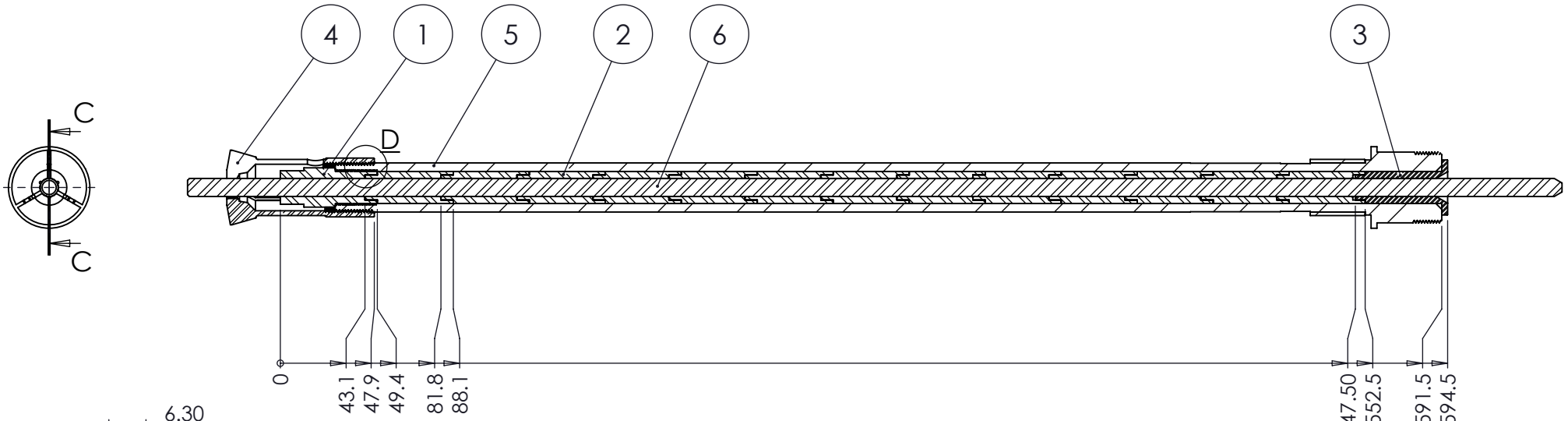
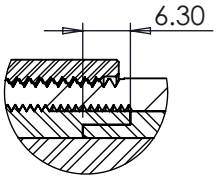
PART No.
TGM-12-363-Bush 13 Pcs.

1 REV.

ITEM NO.	PART NUMBER	PART NAME	DESCRIPTION	QTY.
1	TGM-12-362E-23	Bush For Collect (E)	Made to order	1
2	TGM-12-363C-24	Bush Center (C)	Made to order	13
3	TGM-12-364B-23	Bush (B)	Made to order	1
4	CL-076R-12	Rework collet	Made to order	1
5	TGM-12-406	Draw Tube	Made to order	1
6	Mat'l OD. X.X	Mat'l	G/M	1





DETAIL D
SCALE 1 : 1

SECTION C-C
SCALE 1 : 3

MinebeaMitsumi Inc.
Copy issued Uncontrolled
9/9/2569
Engineering Design center
G/M.1 & M/A. Div.

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. BM-165 | Q'ty 1
MATERIAL -
FINISH -
HEAT TREATMENT -

List Request No. 45/16
3 RD ANGLE
SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 26 Jan' 16

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

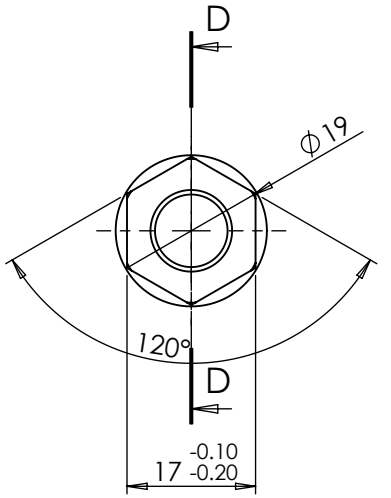
TITLE:
Ass'y Unit Bush Sleeve 15 Pcs.

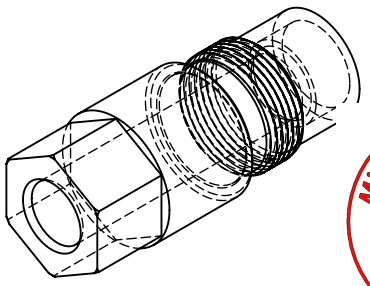
PART No.
TGM-12-363-Bush 13 Pcs.
1 REV.

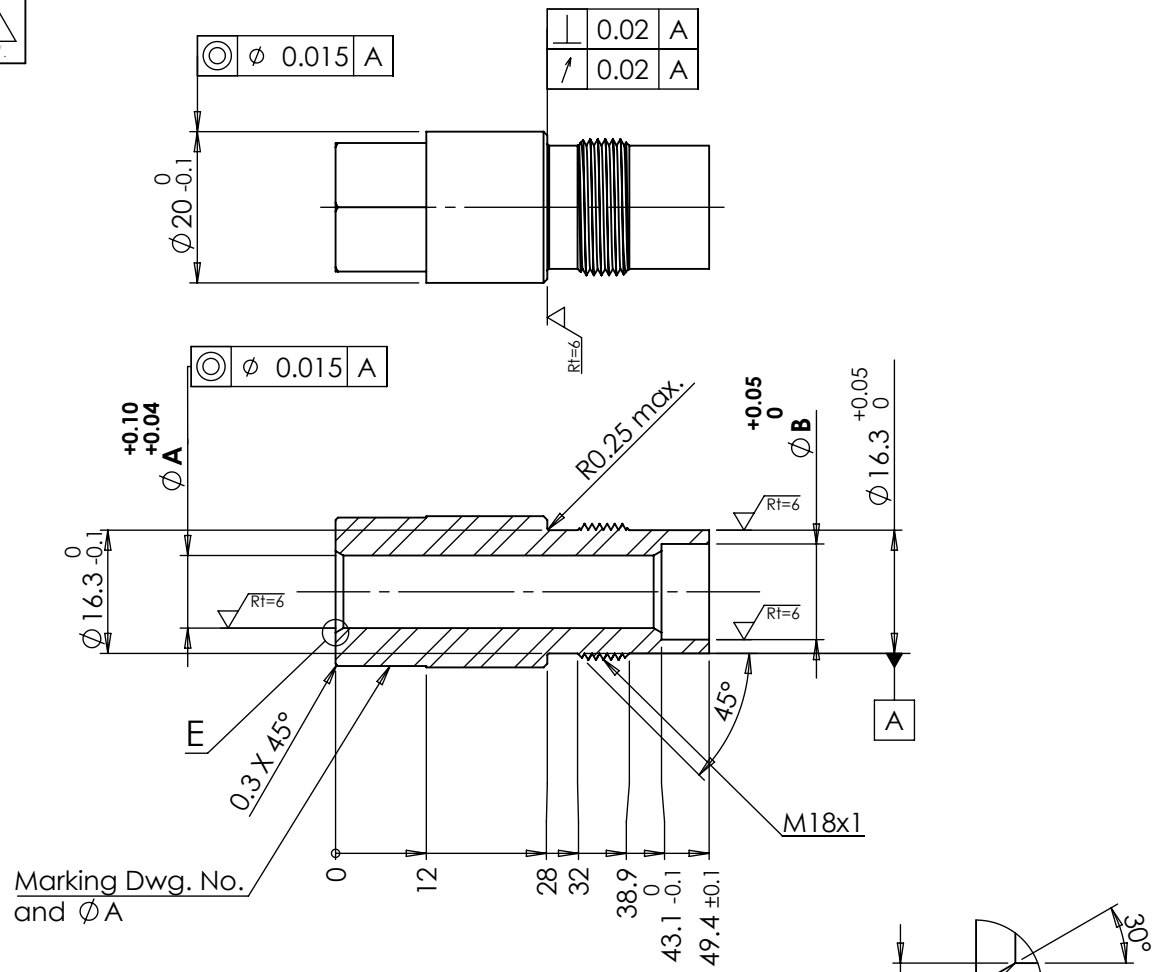
PART No. TGM-12-362E-xx

REV. 1

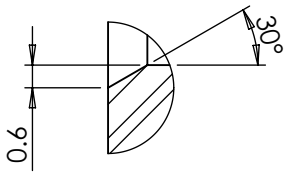
Rt=12 (Rt=6)







SECTION D-D
SCALE 1 : 1



DETAIL E
SCALE 5 : 1
Both Side

MinebeaMitsumi Inc.
Copy issued Uncontrolled
6/10/2569
Engineering Design center
G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Item	⌀A	⌀B	For Mat'l OD.	Check
-01	4.5	10.4	4.5	Preecha
-02	5.12	10.71	5.12	Preecha
-03	6	11.15	6	Preecha
-04	6.1	11.2	6.1	Preecha
-05	6.2	11.25	6.2	Preecha
-06	6.51	11.41	6.51	Preecha
-07	7.2	11.75	7.2	Preecha
-08	7.6	11.95	7.6	Preecha
-09	7.7	12	7.7	Preecha
-10	8.1	12.2	8.1	Preecha
-11	8.2	12.25	8.2	Preecha
-12	9	12.65	9	Preecha
-13	9.2	12.75	9.2	Preecha
-14	9.6	12.95	9.6	Preecha
15	10.8	13.55	10.8	Preecha
-16	11.2	13.75	11.2	Preecha
-17	13.2	14.75	13.2	Preecha
-18	11.5	13.9	11.5	Preecha
-19	12	14.4	12	Preecha
-20	12.2	14.4	12.2	RQ.346/16 Preecha
-21	11	13.55	11	Preecha
-22	10	13.35	10	Preecha
-23	4.4	7.75	4.4	RQ.304/26 Preecha
-24				
-25				

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-165 | Q'ty 1
MATERIAL SUS 303
FINISH -
HEAT TREATMENT -

List Request No. 45/16
3 RD ANGLE
SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Aphisek	Chonknok	Preecha	Preecha

DATE 26 Jan' 16

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Bush For Collect (E)

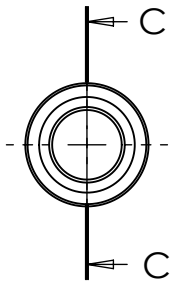
PART No. TGM-12-362E-xx

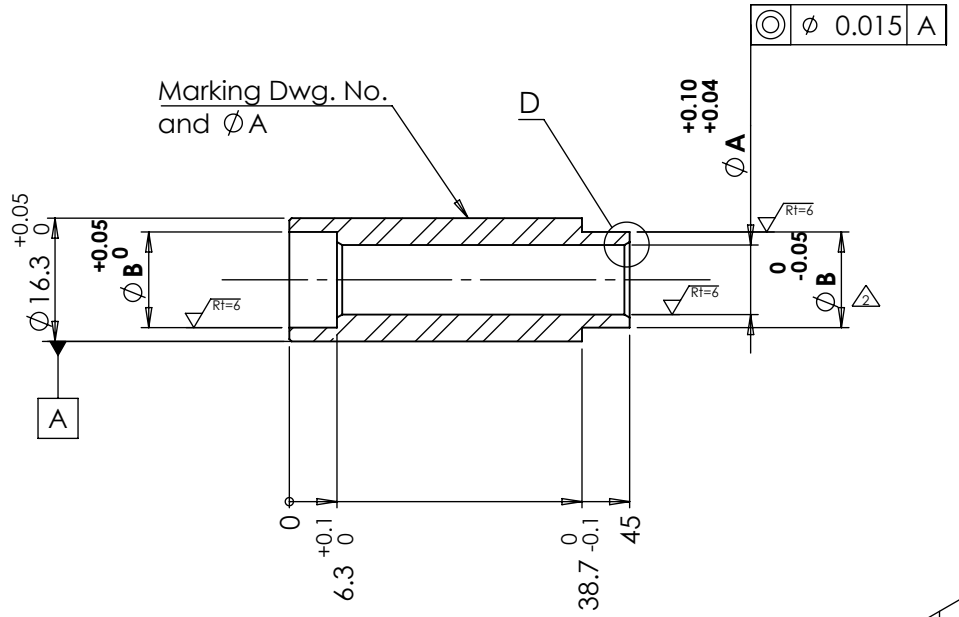
REV. 1

PART No.
TGM-12-363C-xx

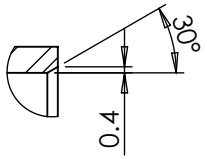
2
REV.

Rt=12 (Rt=6)

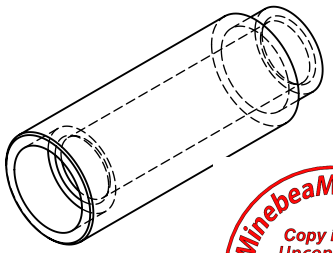




SECTION C-C
SCALE 1 : 1



DETAIL D
SCALE 2 : 1
Both Side



Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

MinebeaMitsumi Inc.
Copy issued Uncontrolled
6/10/2569
Engineering
Design center
G/M.1 & M/A. Div.

Item	ØA	ØB	For Mat'l OD.	Check
-01	4.5	10.4	4.5	Preecha
-02	5.12	10.71	5.12	Preecha
-03	6	11.15	6	Preecha
-04	6.1	11.2	6.1	Preecha
-05	6.2	11.25	6.2	Preecha
-06	6.51	11.41	6.51	Preecha
-07	7.2	11.75	7.2	Preecha
-08	7.6	11.95	7.6	Preecha
-09	7.7	12	7.7	Preecha
-10	8.1	12.2	8.1	Preecha
-11	8.2	12.25	8.2	Preecha
-12	9	12.65	9	Preecha
-13	9.2	12.75	9.2	Preecha
-14	9.6	12.95	9.6	Preecha
15	10.8	13.55	10.8	Preecha
-16	11.2	13.75	11.2	Preecha
-17	13.2	14.75	13.2	Preecha
-18	11.5	13.9	11.5	Preecha
-19	12	14.4	12	Preecha
-20	12.2	14.4	12	Preecha RQ.346/16
-21	13.5	14.9	13.5	Preecha
-22	11	13.55	11	Preecha
-23	10	13.35	10	Preecha RQ.273/24
-24	4.4	7.75	4.4	Preecha RQ.304/26
-25				

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	6 Oct'20	Preecha	Add: ØB

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-165 | Q'ty 13
MATERIAL SUS 303
FINISH -
HEAT TREATMENT -

List Request No. 45/16
3RD ANGLE
SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 26 Jan' 16

Minebea
General Machining Parts Div. 1

TITLE: Bush Center (C)

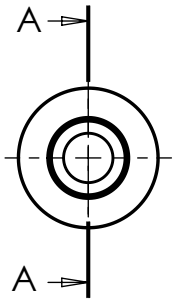
PART No. TGM-12-363C-xx

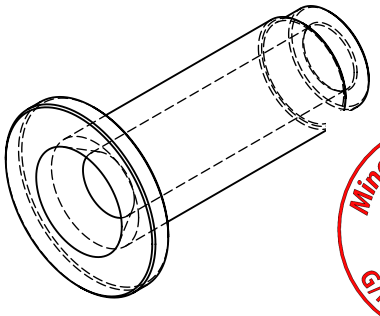
2
REV.

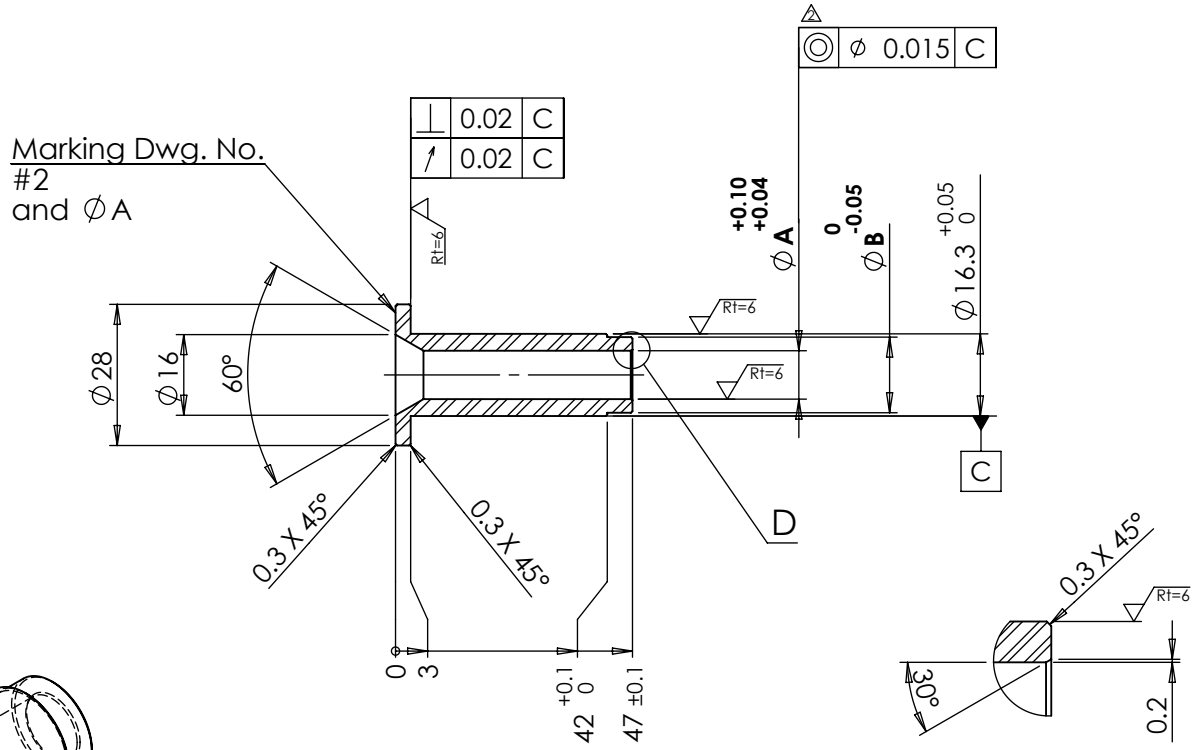
PART No. TGM-12-364B-xx

REV. 2

Rt=12 (Rt=6)







SECTION A-A
SCALE 1 : 1.5

D (2 : 1)

MinebeaMitsumi Inc.
Copy issued Uncontrolled
6/10/2569
Engineering Design center
G/M.1 & M/A. Div.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Item	∅A	∅B	For Mat'l OD.	Check
-01	4.5	10.4	4.5	Preecha
-02	5.12	10.71	5.12	Preecha
-03	6	11.15	6	Preecha
-04	6.1	11.2	6.1	Preecha
-05	6.2	11.25	6.2	Preecha
-06	6.51	11.41	6.51	Preecha
-07	7.2	11.75	7.2	Preecha
-08	7.6	11.95	7.6	Preecha
-09	7.7	12	7.7	Preecha
-10	8.1	12.2	8.1	Preecha
-11	8.2	12.25	8.2	Preecha
-12	9	12.65	9	Preecha
-13	9.2	12.75	9.2	Preecha
-14	9.6	12.95	9.6	Preecha
15	10.8	13.55	10.8	Preecha
-16	11.2	13.75	11.2	Preecha
-17	13.2	14.75	13.2	Preecha
-18	11.5	13.9	11.5	Preecha
-19	12	14.4	12	Preecha
-20	12.2	14.4	12.2	Preecha RQ.346/16
-21	11	13.55	11	Preecha
-22	10	13.35	10	Preecha RQ.273/24
-23	4.4	7.75	4.4	Preecha RQ.304/26
-24				
-25				

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	27 Jan'16	Preecha	Change: ◎ ∅ 0.05 C ==> ◎ ∅ 0.015 C

UNLESS SPECIFIED C0.1 OR R0.1
GENERAL TOLERANCE
JISB0405 MACHING WORKING

LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-165 | Q'ty 1
MATERIAL SUS 303
FINISH -
HEAT TREATMENT -

List Request No. 155/14
3 RD ANGLE
SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 26 Mar' 14

Minebea
General Machining Parts Div. 1

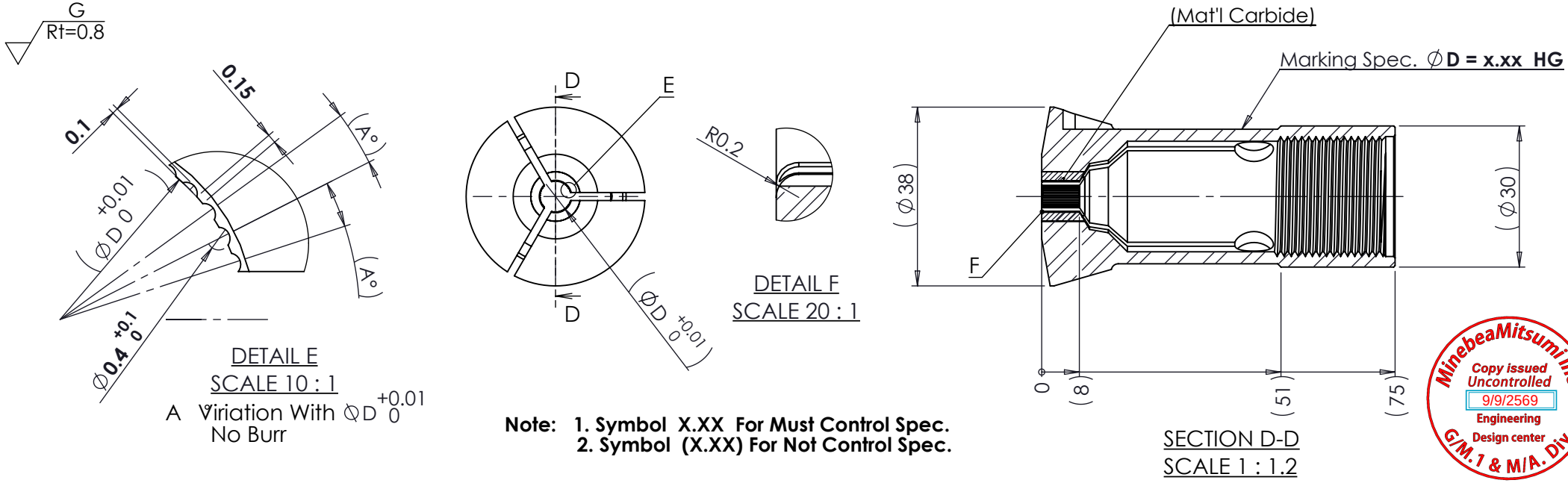
TITLE: Bush (B)

PART No. TGM-12-364B-xx

REV. 2

PART No. CL-078R-xx

1 REV. **Rework Groove only**



Note: 1. Symbol X.XX For Must Control Spec.
2. Symbol (X.XX) For Not Control Spec.

MinebeaMitsumi Inc. Copy issued Uncontrolled 9/9/2569 Engineering Design center G/M.1 & M/A. Div.

Item	(ØD)	Check
-01	11.2	Preecha
-02	6.1	Preecha
-03	8.1	Preecha
-04	9.6	Preecha
-05	8.3	Preecha
-06	6.2	Preecha
-07	9.2	Preecha
-08	9	Preecha
-09	5.12	Preecha
-10	7.7	Preecha

Item	(ØD)	Check
-11	11.5	Preecha
-12	13.2	Preecha
-13	12	Preecha
-14	6.51	Preecha
-15	9.7	Preecha
-16	10	Preecha RQ.472/20
-17	11	Preecha RQ.472/20
-18	16.5	Preecha RQ.88/21
-19	17	Preecha RQ.88/21
-20	10.3	Preecha RQ.369/22

Item	(ØD)	Check
-21	13	Preecha RQ.369/22
-22	4.4	Preecha RQ.269/26
-23		
-24		
-25		
-26		
-27		
-28		
-29		
-30		

Item	(ØD)	Check
-31		
-32		
-33		
-34		
-35		
-36		
-37		
-38		
-39		
-40		

Item	(ØD)	Check
-41		
-42		
-43		
-44		
-45		
-46		
-47		
-48		
-49		
-50		

Item	(ØD)	Check
-51		
-52		
-53		
-54		
-55		
-56		
-57		
-58		
-59		
-60		

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S ~ UNCUT

M/C. BM-16 BM-165 | Q'ty 1

MATERIAL (SCM, Carbide)

FINISH -

HEAT TREATMENT -

List Request No.

3 RD ANGLE

SCALE : | SHEET / | UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 18 Nov' 09

NMB MinebeaMitsumi Inc. *General Machining Parts Div. 1*

TITLE: Rework High grip Collet

PART No. CL-078R-xx

1 REV.

PART No.

TGM-12-406

3 REV.

Note: 1. Break all Corner approx by C0.2
2. General Tolerance ±0.1

Main Spindle

REV.	ECN No.	DATE	NAME	DESCRIPTION
1	----	----	---	ENGINEERING RELEASE
2	----	22 Jan'14	Preecha	Change: Ø16.5 ==> 16.3 etc, RQ33/14
3	----	10 Feb'17	Preecha	Add: Bush etc, RQ.81/17

UNLESS SPECIFIED C0.1 OR R0.1	
GENERAL TOLERANCE	
JISB0405 MACHING WORKING	
LESS THAN 0.5	±0.05
0.5 ~ 6	±0.1
6 ~ 30	±0.2
30 ~ 120	±0.3
120 ~ 400	±0.4
400 ~ 1000	±0.8
ANGLE	±1°

0.8S 6.3S 25S 100S UNCUT

M/C. BM-16, BM165 Q'ty 1

MATERIAL STKM, SKS-3

FINISH Demagnetize Max0.3T (3G)

HEAT TREATMENT Black Oxide

List Request No. 1686

3 RD ANGLE

SCALE : SHEET / UNIT (mm.)

DETAILED	DESIGNED	CHECKED	APPROVED
Phalachai	Aphisek	Preecha	Preecha

DATE 16 Sep' 16

NMB MinebeaMitsumi Inc.
General Machining Parts Div. 1

TITLE: Draw Tube

PART No. TGM-12-406

3 REV.

MinebeaMitsumi Inc.
Copy issued Uncontrolled
9/9/2569
Engineering Design center
G/M.1 & M/A. Div.